

PRASHANTMAURYA

Backend Engineer | Java, Spring Boot, AI-Integrated Systems

+91-8057971683 | mauryaprashant1111@gmail.com | [GitHub](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

Results-driven Backend Engineer with hands-on experience building production-grade microservices and AI-integrated systems using Java (Spring Boot) and Python. Proven track record designing REST APIs, JWT-secured architectures, semantic search pipelines, and agentic LLM workflows. Passionate about bridging backend engineering and applied AI to deliver reliable, scalable systems.

TECHNICAL SKILLS

Languages:	Java, Python
Backend:	Spring Boot, FastAPI, REST APIs, Microservices, MVC, Layered Architecture
AI / ML:	LangChain, LangGraph, Sentence Transformers, Semantic Search, RAG, Agentic AI
Security:	Spring Security, JWT Authentication
Databases:	MySQL, PostgreSQL
Tools & Other:	Git, Postman, React, Vite

PROJECTS

Diagnyx — Agentic AI Medical Report Interpretation System (diagnyx.netlify.app/)

FastAPI · LangGraph · LangChain · Groq LLM · OCR · Pydantic

- Architected a 9-stage agentic AI pipeline (ingestion → reasoning → synthesis) for automated medical report analysis using LangGraph state machines.
- Engineered multi-step LLM workflow with iterative feedback loops, reducing hallucination rate through rule-based safety validation against clinical thresholds.
- Implemented multi-format document ingestion (PDF, DOCX, OCR images) with preprocessing pipeline; used structured Pydantic schemas for consistent API responses.

Lost & Found — AI-Powered Semantic Matching Microservices System

Spring Boot · FastAPI · Sentence Transformers · MySQL · JWT · REST

- Designed dual-service architecture (Spring Boot + FastAPI) with REST inter-service communication, achieving clean separation of business logic and AI inference layers.
- Built semantic search engine using 384-dimensional sentence embeddings and cosine similarity; developed hybrid ranking algorithm (semantic + category + fuzzy location) achieving **85% match accuracy**.
- Implemented JWT-based authentication with role-based access control (RBAC) and real-time matching pipeline with automated email notifications.

BloodConnect — Geospatial Blood Donation Platform Backend (bloodconnect-app-eight.vercel.app/)

Spring Boot · PostgreSQL · JWT · Spring Security · Haversine · React

- Built RESTful backend with role-based access control (HOSPITAL / DONOR roles), enforcing blood compatibility rules and 56-day donation cooldown business logic.
- Implemented geospatial donor discovery using the Haversine formula for proximity-based matching; designed automated notification system for real-time donor alerts.
- Secured platform with JWT authentication and Spring Security; integrated React/Vite frontend consuming REST APIs.

EDUCATION

B.Tech — Computer Science & Engineering

2023 – 2027

Babu Banarasi Das University, Lucknow